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(54) QUICK DISCONNECT TERMINAL WITH SPLIT MATING INTERFACE

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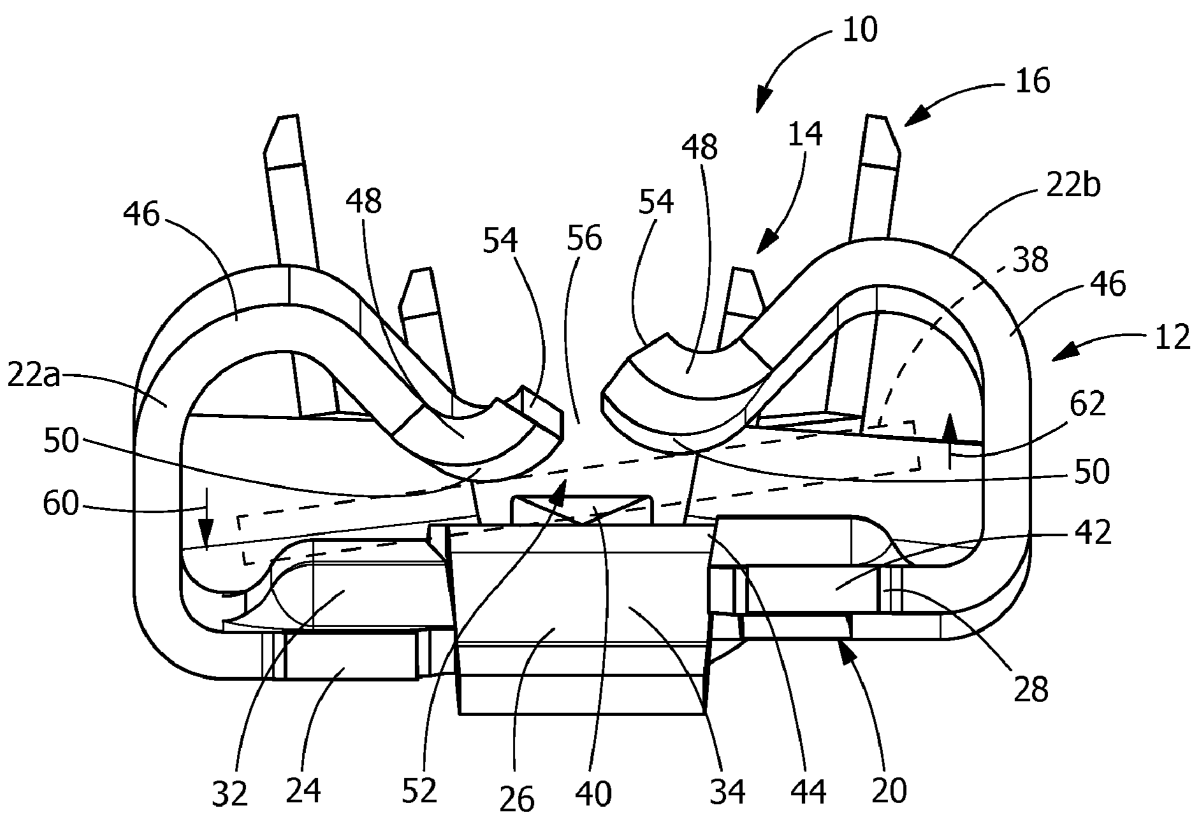
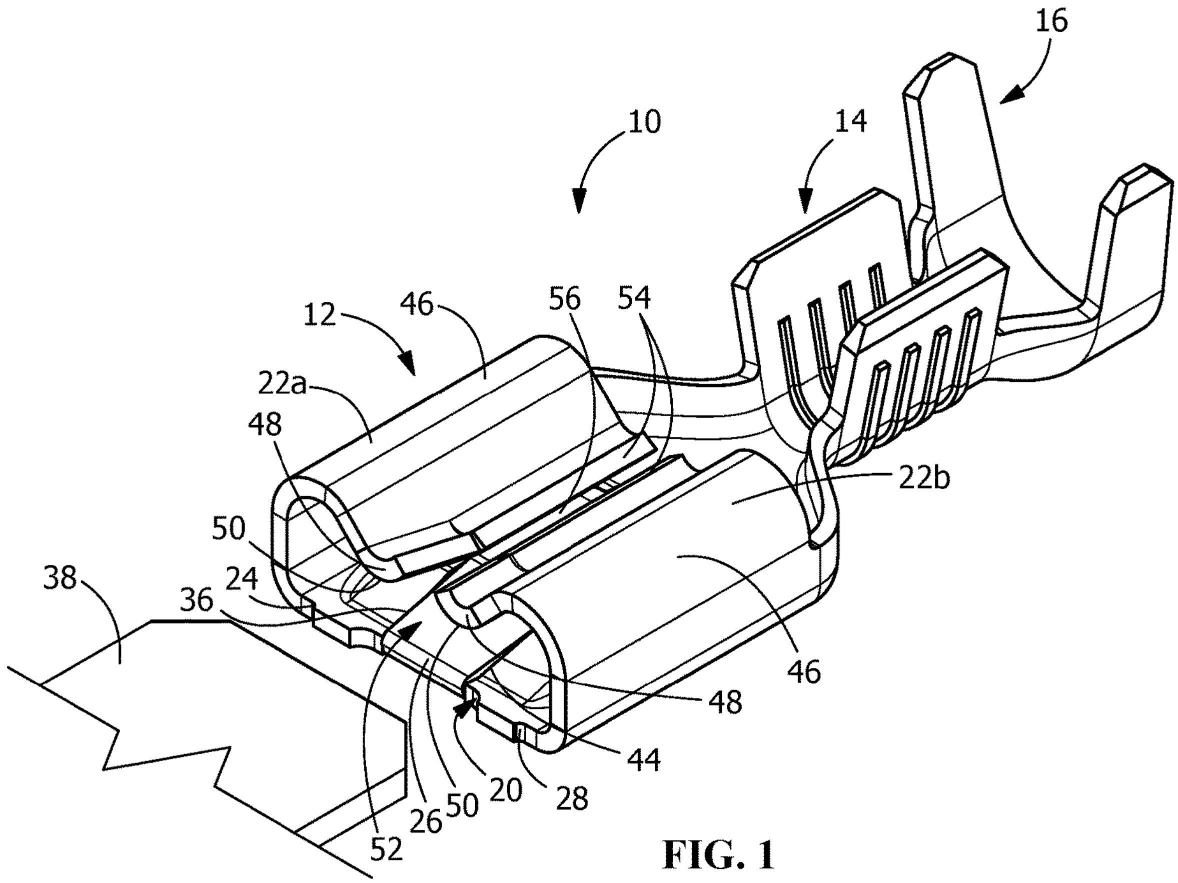
(57) **ABSTRACT**

An electrical terminal with a contact portion that has a bottom wall with resilient contact arms extending therefrom. A mating slot is positioned between the resilient contact and the bottom wall. The bottom wall has a split mating interface with a first section, a second section and a third section. The second section is separated from the first section by a first slot. The third section separated from the second section by a second slot. The first section, the second section and the third section move independently relative to each other in a direction perpendicular to an axis of insertion of a mating terminal to provide and maintain a stable and reliable electrical connection between a mating terminal inserted into the mating slot and the terminal.




16
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28

FIG. 1

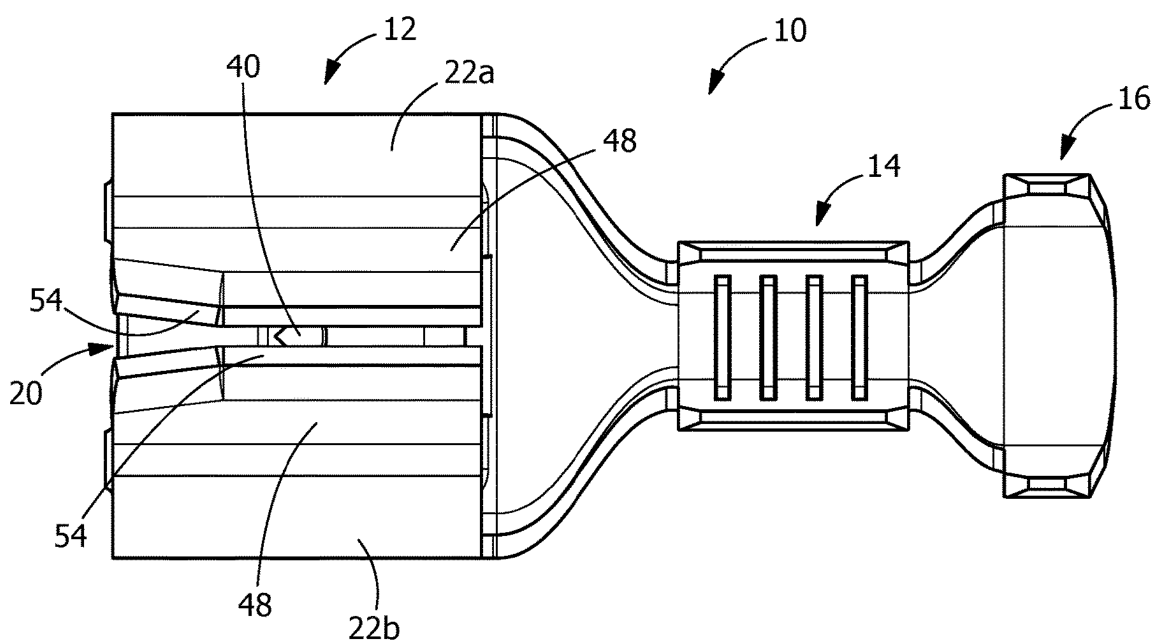
12
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FIG. 2

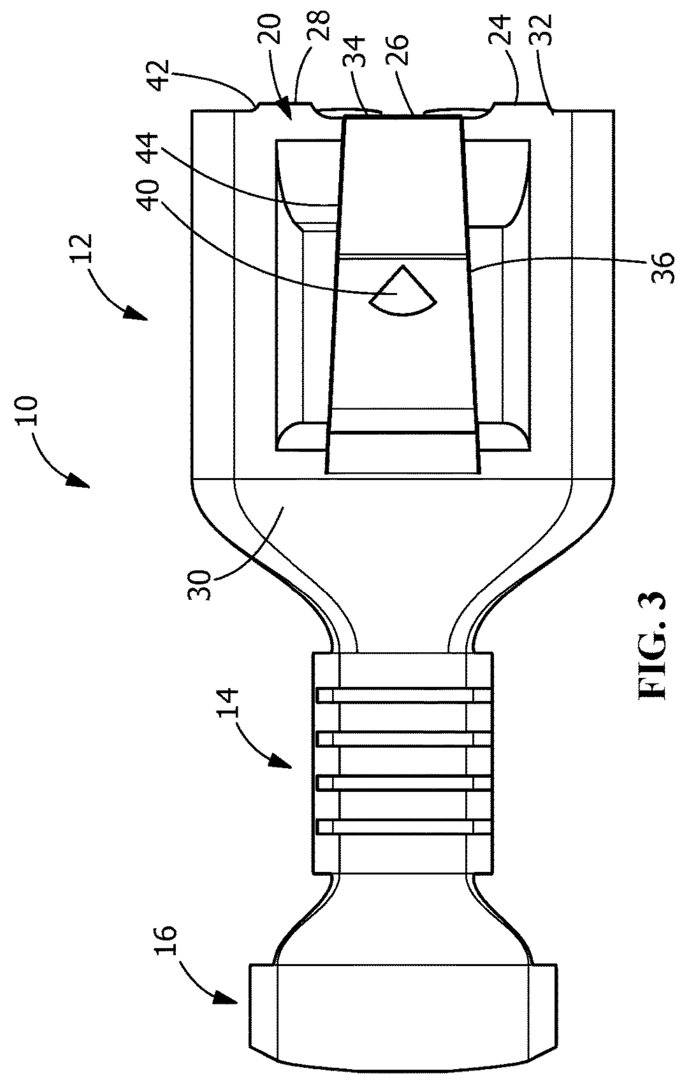

FIG. 3

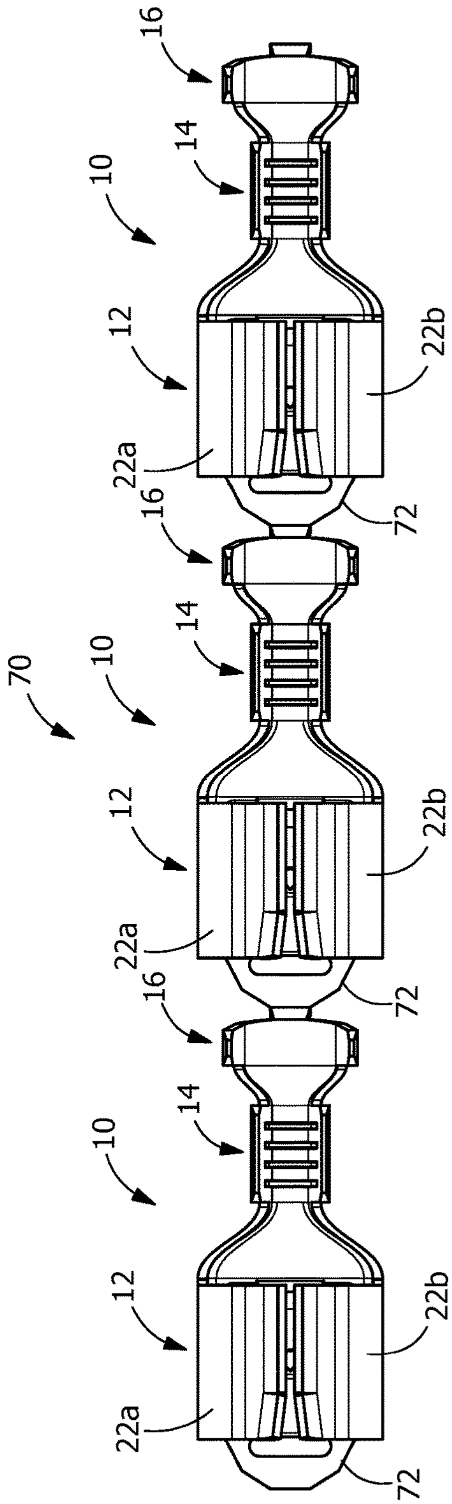

FIG. 6

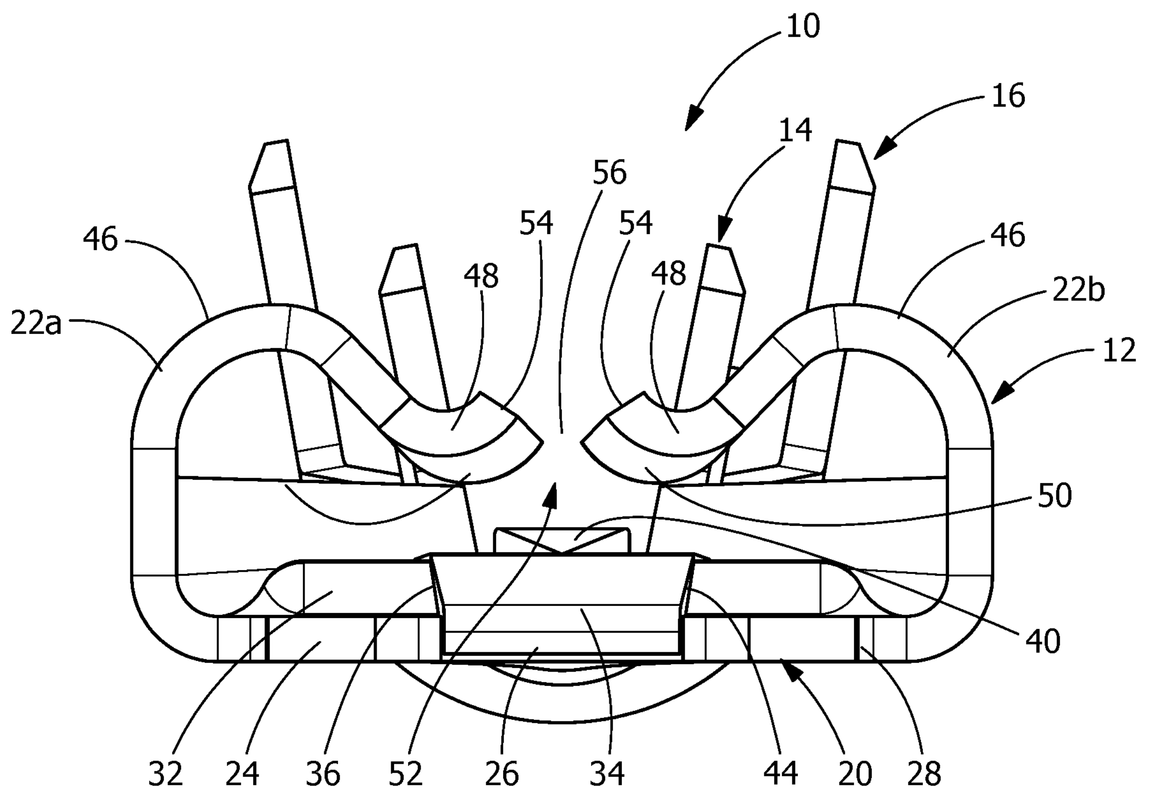
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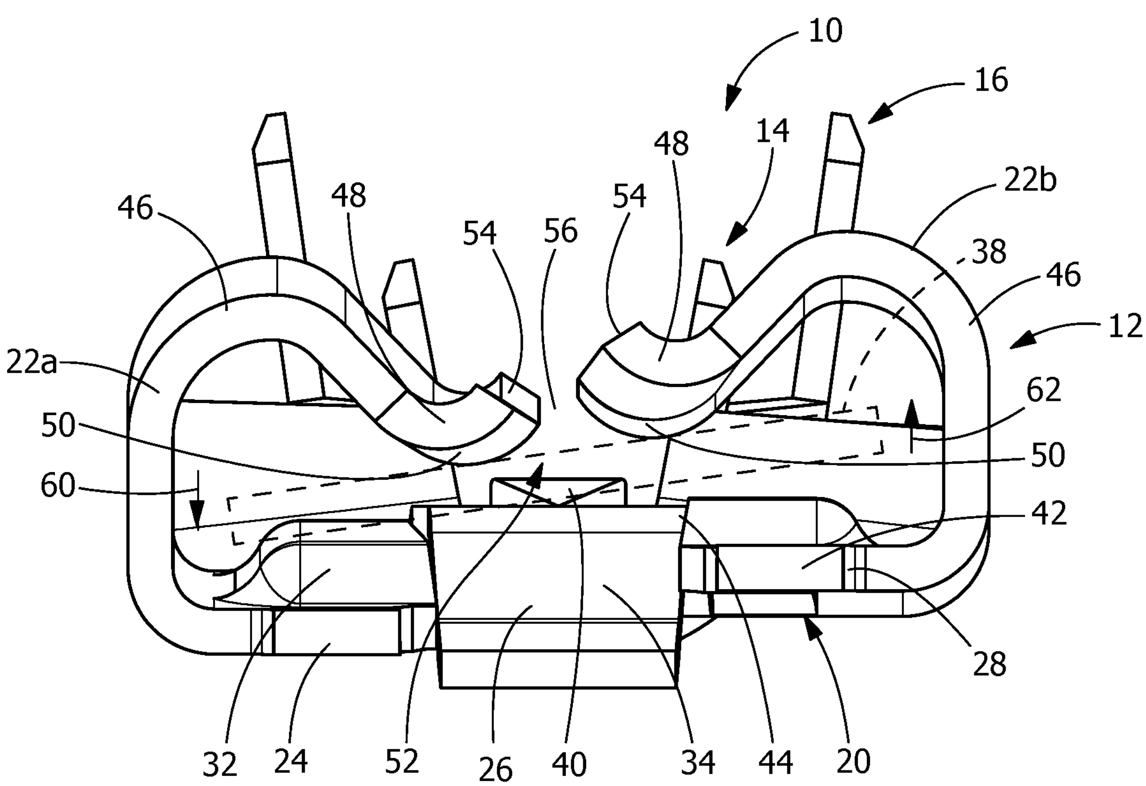
FIG. 4


10
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22b
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20

FIG. 5

QUICK DISCONNECT TERMINAL WITH SPLIT MATING INTERFACE

FIELD OF THE INVENTION

[0001] The present invention is directed to a quick connect electrical terminal. In particular, the invention is directed to a receptacle which minimizes stubbing of a mating electrical terminal and which provides a stable electrical connection even when the mating terminal is misaligned.

BACKGROUND OF THE INVENTION

[0002] Quick connect electrical terminals, such as tab receptacle terminals, which are adapted for quick make and break connections with a mating terminal or mating tab, are known. Terminals of this kind are used to make an electrical connection to a male or tab terminal which is inserted and held in the socket terminal.

[0003] It is often necessary to disconnect and reconnect such terminals a number of times, for example, for testing purposes prior to final inspection and shipment of the product on which such terminals are used. It is also required that the connection made with such terminals be maintained under conditions of vibration and possible strain in subsequent service. Traditionally, these terminals have limited flexibility or movement which causes stubbing and high insertion forces. In addition, the contact areas of the terminals may be imbalanced resulting in an unstable electrical connection and a lesser current carrying capability.

[0004] It would, therefore, be beneficial to provide a quick connect receptacle terminal in which the mating interface provides sufficient flexibility to avoid stubbing and potential damage to the mating terminal. In addition, it would be beneficial to provide a receptacle terminal in which the mating interface is allowed to move with the mating terminal to provide a more stable electrical connection.

SUMMARY OF THE INVENTION

[0005] An embodiment is directed to an electrical terminal with a contact portion that has a bottom wall with resilient contact arms extending therefrom. The bottom wall has a first section, a second section and a third section. The first section is separated from the second section by a first slot. The second section is separated from the third section by a second slot. The first section, the second section and the third section move independently relative to each other in a direction perpendicular to an axis of insertion of a mating terminal.

[0006] An embodiment is directed to an electrical terminal with a contact portion that has a bottom wall with resilient contact arms extending therefrom. A mating slot is positioned between the resilient contact and the bottom wall. The bottom wall has a split mating interface with a first section, a second section and a third section. The second section is separated from the first section by a first slot. The third section separated from the second section by a second slot. The first section, the second section and the third section move independently relative to each other in a direction perpendicular to an axis of insertion of a mating terminal to provide and maintain a stable and reliable electrical connection between a mating terminal inserted into the mating slot and the terminal.

[0007] Other features and advantages of the present invention will be apparent from the following more detailed description of the preferred embodiment, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is perspective view of an illustrative embodiment of a receptacle terminal according to the present invention, a mating tab terminal is shown prior to being inserted into the receptacle terminal.

[0009] FIG. **2** is a top view of the receptacle terminal of FIG. **1**.

[0010] FIG. **3** is a bottom view of the receptacle terminal of FIG. **1**.

[0011] FIG. **4** is a front view of the receptacle terminal of FIG. **1** prior to the mating tab terminal being inserted into a terminal receiving area of the receptacle terminal.

[0012] FIG. **5** is a front view of the receptacle terminal of FIG. **1**, with the mating tab terminal inserted into the terminal receiving area of the receptacle terminal, the tab terminal is slightly misaligned with the receptacle terminal, causing the mating interface of the receptacle terminal to be flexed to accommodate the tab terminal.

[0013] FIG. **6** top view of a several receptacle terminals shown in a strip.

DETAILED DESCRIPTION OF THE INVENTION

[0014] The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as "lower," "upper," "horizontal," "vertical," "above," "below," "up," "down," "top" and "bottom" as well as derivative thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as "attached," "affixed," "connected," "coupled," "interconnected," and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

[0015] Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features, the scope of the invention being defined by the claims appended hereto.

[0016] As best shown in FIGS. **1** through **5**, a quick disconnect electrical receptacle, socket or female terminal **10** includes a contact portion **12**, a wire barrel portion **14** behind the contact portion **12** and an insulation barrel portion **16** behind the wire barrel portion **14**. The wire barrel portion **14** is configured for a crimped connection with an end of a conductive core of an insulated wire. The insulation barrel portion **16** is configured for a crimped connection with an end of the insulation coating or jacket of the wire. Although a wire barrel portion **14** and an insulation barrel portion **16** are shown, the contact portion **12** can be used with other types of termination members without departing from the scope of the invention. In the illustrative embodiment shown, the terminal **10** is stamped and formed from a metal plate having good electrical conductivity.

[0017] The contact portion **12** includes a bottom wall **20** and resilient contact arms **22***a*, **22***b* which extend from either side of the bottom wall **20**. As shown in FIG. **3**, the bottom wall **20** has a first section **24**, a second section or spring arm **26** and a third section **28**. The first section **24**, the second section **26** and the third section **28** extend from, and are integrally attached to, a base section **30** of the bottom wall **20**. The base section **30** is provided proximate the wire barrel portion **14**. The first section **24**, the second section **26** and the third section **28** form a split beam mating interface.

[0018] The first section **24** extends from base section **30** to a free end **32**. The resilient contact arm **22***a* extends from the side of the first section **24**.

[0019] The second section **26** extends from base section **30** to a free end **34**. The second section **26** is separated from the first section **24** by a slot or gap **36**. The slot or gap **36** allows the free end **34** of the second section **26** to move independently relative to the free end **32** of the first section **24** in a direction perpendicular to the axis of insertion of a mating terminal **38** (FIGS. **1** and **5**) in the contact portion **12** of the terminal **10**. The second section **26** has a projection or embossment **40**, such as, but not limited to, a projection, detent, dimple or lance which is formed from the second section **26** to create a raised area on an inner surface of the second section **26**. The projection or embossment **40** engages the mating terminal **38** as the mating terminal **38** is inserted into the terminal **10**.

[0020] The third section **28** extends from base section **30** to a free end **42**. The resilient contact arm **22***b* extends from the side of the third section **28**. The third section **28** is separated from the second section **26** by a slot or gap **44**. The slot or gap **44** allows the free end **42** of the third section **28** to move independently relative to the free end **34** of the second section **26** and relative to the free end **32** of the first section **24** in a direction perpendicular to the axis of insertion of the mating terminal **38** (FIGS. **1** and **5**) in the contact portion **12** of the terminal **10**.

[0021] In the illustrative embodiment shown in FIG. **1**, each resilient arm **22***a*, **22***b* has a resilient contact section **46** with an arcuate or curled portion which extends from the bottom wall **20** to a mating terminal engaging member **48**. The configuration of the resilient contact arms **22***a*, **22***b* allows the stiffness and spring rate of the resilient contact arms **22***a*, **22***b* to be controlled.

[0022] The mating terminal engagement members **48** extend from the resilient contact section **46**. A mating terminal engagement surface **50** is provided on each mating terminal engaging member **48**. In the embodiment shown, the mating terminal engaging member **48** extends from the resilient contact arms **22***a*, **22***b*, positioning the mating terminal engagement surface **50** at the top of a terminal mating slot **52**. The configuration of the resilient contact arms **22***a*, **22***b* provides resiliency to allow the mating terminal engaging member **48** to move relative to the bottom wall **20** as the mating terminal is inserted into the mating slot **52**. The mating terminal engagement surfaces **50** have an arcuate or rounded configuration. However, other configurations of the engagement surfaces **50** may be used.

[0023] The mating terminal engagement members **48** have end surfaces **54** which are positioned proximate to each other, but are spaced apart to form a slot **56**. The width of the slot **56** is large enough to allow the end surfaces **54** and their respective contact arms **22***a*, **22***b* to move independently of each other. However, the width of the slot **56** is minimized to increase the beam length of the contact arms **22***a*, **22***b*. The beam length allows the contact arms **22***a*, **22***b* to have enhanced resilient or spring characteristics, allowing the contact arms **22***a*, **22***b* to move without taking a permanent set, failing or yielding. The slot **56** is positioned in line with the projection or embossment **40** of the second portion **26** of the bottom wall **20**.

[0024] As shown in FIG. **4**, in the initial, unstressed position, prior to the insertion of the mating terminal **38** into the mating slot **52**, the embossment **40** of the second section or spring arm **26** is positioned proximate to, but not in engagement with, the mating terminal engagement surfaces **50** of the mating terminal engagement members **48**. In this unstressed position, the resilient arm **22***a* and the resilient arm **22***b* are essentially mirror images of each other. The mating terminal engagement surfaces **50** of the mating terminal engagement members **48** are positioned in essentially the same plane which is transverse to the longitudinal axis of the terminal **10**.

[0025] In the unstressed position, the mating terminal engagement surfaces **50** of the mating terminal engagement members **48** and the second section or spring arm **26** defined the mating slot **52**. In this position, the resilient arms **22***a*, **22***b* are in an unstressed position. Similarly, the first section **24**, the second section or spring arm **26** and the third section **28** are also in an unstressed position, allowing the free end **32** of the first section **24**, the free end **34** of the second section **26**, and the free end **42** of the third section **28** to be positioned essentially in line with each other.

[0026] As previously described, the slot **36** and the slot **44** provided in the bottom wall **20** allow the first section **24**, the second section or spring arm **26** and the third section **28** to move independently of each other. Consequently, during insertion of the mating terminal **38** into the mating slot **52** of the contact portion **12** of the terminal **10**, the contact portion **12**, including the split mating interface of the first section **24**, the second section **26** and the third section **28**, may conform or adapt to a mating terminal **38** which is misaligned during mating, as shown in FIG. **5**.

[0027] As shown in the illustrative embodiment of FIG. **5**, the first section **24** and the resilient arm **22***a* attached thereto are moved downward, as represented by arrow **60**, relative to the second section **26**. In addition, the third section **28** and the resilient arm **22***b* attached thereto are moved upward, as represented by arrow **62**, relative to the second section **26**. This allows the contact section **12** to accommodate a mating terminal **38** which is misaligned in the manner shown in FIG. **5**. However, if the orientation of the misaligned mating terminal **38** is different, the first section **24** and the resilient arm **22***a* may be moved upward and the third section **28** and the resilient arm **22***b* may be moved downward.

[0028] The relative independent movement of the first section **24**, the second section or spring arm **26** and the third section **28** allow for the contact portion **12** of the terminal **10** to compensate for any slight misalignment of the mating terminal **38** or any slight warpage or imperfections associated with the mating terminal **38**.

[0029] In a fully inserted position, the mating terminal engagement surface **50** of each of the resilient arms **22***a*, **22***b* are provided and maintained in physical and electrical engagement with the mating terminal **38**. Due to the independent movement of the first section **24** and the third section **28**, multiple points of contact are made and maintained between each mating terminal engagement surface **50** and the mating terminal **38**. In addition, the embossment **40** or the second section or spring arm **26** is also provided and maintained in electrical and mechanical contact with the mating terminal **38**. The multiple areas of contact allow the terminal **10** to be used in applications with higher current levels, such as, but not limited to, 15 to 20 or more amps.

[0030] The independent movement of the first section **24**, the second section or spring arm **26** and the third section **28** provide and maintains a stable and reliable electrical connection between the mating terminal **38** and the terminal **10**. The independent movement of the first section **24**, the second section or spring arm **26** and the third section **28** also allows for higher hertzian stresses when the mating terminal **38** is mated to the terminal **10**, thereby eliminating or minimizing the fretting corrosion between the terminal **10** and the mating terminal **38**, further contributing to a stable and reliable electrical connection between the mating terminal **38** and the terminal **10**.

[0031] Portions of the mating terminal engagement surfaces **50** are spaced from and are laterally offset from the embossment **40**, allowing the engagement between the mating terminal **38** and the mating terminal engagement surfaces **50** and the embossment **40** of the terminal **10** to be dispersed, i.e. not at one point or in a straight line. This provides a stable mechanical and electrical engagement between the mating terminal **38** and receptacle terminal **10** in all environments, thereby insuring that the mating terminal **38** will remain properly positioned in the receptacle terminal **10** as vibration occurs or if the mating terminal **38** is bent.

[0032] A terminal according to the teaching of the invention has a lower spring rate than known terminals. As the first section **24**, the second section or spring arm **26** and the third section **28** move independently, the resilient arms **22***a*, **22***b* can have a reduced spring rate while still providing sufficient normal force to retain the mating terminal **38** in the mating slot **52** of the contact portion **12** of the terminal **10**. By controlling the spring rate of the spring arms **22***a*, **22***b*, the normal forces and insertion forces associated with the contact portion **12** of the terminal **10** can also be controlled, while allowing for a proper electrical connection between the terminals **10** and the mating terminals. For example, the insertion force of a terminal **10** made according to the present invention may be reduced in comparison to a terminal which has the first section **24**, the second section or spring arm **26** and the third section **28** fixed to each other.

[0033] As the spring rate is reduced, the resilient arms **22***a*, **22***b* allow for a greater spring deflection before yielding or taking a permanent set. This allows the terminal **10** to be used with mating terminals **38** which have some variance in manufacturing tolerances. In other words, because the resilient arms **22***a*, **22***b* have the ability to deflect a greater distance without yielding or taking a permanent set, the thickness of the mating terminal **38** does not have be as precisely controlled. This is advantageous with terminals made of materials, such as steel, which do not exhibit generally poor spring characteristics.

[0034] Referring to FIG. **6**, a portion of an illustrative strip **70** of terminals **10** are shown. A rigid front bar **72** is provide between each terminal **10** on the strip **10**. The rigid front bar **72** is stamped and formed as the terminals **10** are stamped and formed. The rigid front bar **72** prevent damage to the terminals **10** during the reeling, plating and shipping of the terminals **10** and strip **70**. The rigid front bar **72** is integrally attached to the bottom wall **20**. The rigid front bar **72** extends from the first section **24** to the third section **28** of the bottom wall **20**. The rigid front bar **72** cooperates with the first section **24** and the third section **28** to maintain the first section **24** and the third section **28** in a relatively fixed position relative to each other.

[0035] In the illustrative embodiment shown, the rigid front bar **72** is sheared off when a terminal **10** is terminated or crimped to a wire or cable. With the rigid front bar **72** removed the first section **24**, second section **26** and third section **28** are allowed to move relative to each other as described above.

[0036] While the invention has been described with reference to a preferred embodiment, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the spirit and scope of the invention as defined in the accompanying claims. One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials and components and otherwise used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims, and not limited to the foregoing description or embodiments.

1. An electrical terminal comprising:

a contact portion having a bottom wall with resilient contact arms extending therefrom;

the bottom wall having a first section, a second section and a third section, the first section separated from the second section by a first slot, the second section separated from the third section by a second slot;

wherein the first section, the second section and the third section move independently relative to one another in a direction perpendicular to an axis of insertion of a mating terminal.

2. The electrical terminal as recited in claim **1**, wherein a first resilient contact arm of the resilient contact arms extends from the first section of the bottom wall and a second resilient contact arm of the resilient contact arms extends from the third section of the bottom wall.

3. The electrical terminal as recited in claim **1**, wherein the second section is a spring arm.

4. The electrical terminal as recited in claim **3**, wherein an embossment is provided on the second section.

5. The electrical terminal as recited in claim **1**, wherein the first section, the second section and the third section extend from, and are integrally attached to a base section of the bottom wall.

6. The electrical terminal as recited in claim **1**, wherein the first section extends from a base section of the bottom wall to a first free end, a first resilient contact arm extends from the first section.

7. The electrical terminal as recited in claim **6**, wherein the second section extends from the base section to a second free end, the second free end of the second section moves independently relative to the first free end of the first section in a direction perpendicular to the axis of insertion of the mating terminal, the second section has an embossment formed from the second section to create a raised area on an inner surface of the second section.

8. The electrical terminal as recited in claim **7**, wherein the third section extends from the base section to a third free end, a second resilient contact arm extends from the third section, the third free end of the third section moves independently relative to the first free end of the first section and the second free end of the second section in a direction perpendicular to the axis of insertion of the mating terminal.

9. The electrical terminal as recited in claim **8**, wherein the first resilient arm and the second resilient arm have resilient contact sections with arcuate portions which extend from the bottom wall to mating terminal engaging members.

10. The electrical terminal as recited in claim **9**, wherein mating terminal engagement surfaces are provided on the mating terminal engaging members, the mating terminal engagement surfaces are positioned at a top of a terminal mating slot.

11. The electrical terminal as recited in claim **10**, wherein the mating terminal engagement members have end surfaces which are positioned proximate to each other, but are spaced apart to form a slot, the ends surfaces and their respective contact arms move independently of each other.

12. The electrical terminal as recited in claim **1**, wherein the contact portion is moveable between a first position and a second position, in the first position, prior to the insertion of the mating terminal into a mating slot of the contact portion, the contact portion is in an unstressed position, in the second position, with the mating terminal inserted into the mating slot of the contact portion, the contact portion is in a stressed position.

13. An electrical terminal comprising:

a contact portion having a bottom wall with resilient contact arms extending therefrom, a mating slot positioned between the resilient contact and the bottom wall;

the bottom wall having a split mating interface comprising:

a first section;

a second section separated from the first section by a first slot;

a third section separated from the second section by a second slot;

wherein the first section, the second section and the third section move independently relative to one another in a direction perpendicular to an axis of insertion of a mating terminal to provide and maintain a stable and reliable electrical connection between a mating terminal inserted into the mating slot and the terminal.

14. The electrical terminal as recited in claim **13**, wherein a first resilient contact arm of the resilient contact arms extends from the first section of the bottom wall and a second resilient contact arm of the resilient contact arms extends from the third section of the bottom wall.

15. The electrical terminal as recited in claim **13**, wherein the second section is a spring arm.

16. The electrical terminal as recited in claim **13**, wherein the first section, the second section and the third section extend from, and are integrally attached to a base section of the bottom wall.

17. The electrical terminal as recited in claim **13**, wherein the first section extends from a base section of the bottom wall to a first free end, a first resilient contact arm extends from the first section.

18. The electrical terminal as recited in claim **17**, wherein the second section extends from the base section to a second free end, the second free end of the second section moves independently relative to the first free end of the first section in a direction perpendicular to the axis of insertion of the mating terminal, the second section has an embossment formed from the second section to create a raised area on an inner surface of the second section.

19. The electrical terminal as recited in claim **18**, wherein the third section extends from the base section to a third free end, a second resilient contact arm extends from the third section, the third free end of the third section moves independently relative to the first free end of the first section and the second free end of the second section in a direction perpendicular to the axis of insertion of the mating terminal.

20. The electrical terminal as recited in claim **19**, wherein the first resilient arm and the second resilient arm have resilient contact sections with arcuate portions which extend from the bottom wall to mating terminal engaging members.

* * * * *